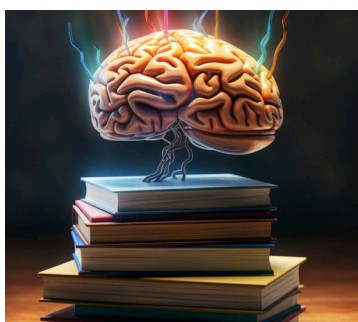


Generative AI Platforms: A Boon for the Academics and Research Communities

Classical teaching methods are poised to go through a total transformation with the advent of artificial intelligence. AI-based tools are already helping teachers, students and researchers to create accurate content and enhance their knowledge.

Artificial intelligence-based applications are being used these days by most professionals for writing content, editing, data analytics, and other routine office practices. The academic world is also making use of AI-based software applications for getting accurate content rather than using classical search engines.



The global market for AI-based robotics applications is increasing at an exponential speed. Many chatbots are today available for multiple professional applications including data engineering, content evaluation, speech recognition, text analysis, and so on.

Artificial Intelligence (AI) driven robots market size worldwide in 2021 with a forecast until 2030
(in million U.S. dollars) Source : Statista.com

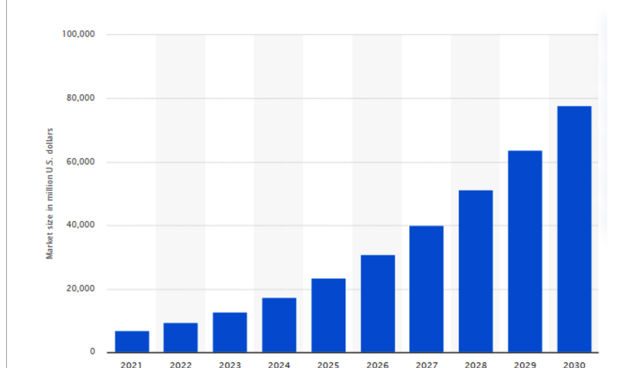


Figure 1: Global market size of AI-driven robots

In academics, multiple types of content are required on a regular basis and these can be generated using AI. Earlier, such content was generated manually by teachers and faculty members. With the advent of AI-based applications, routine tasks can be effectively automated. In addition, teaching content is generated by AI in a professional way and hence can be accepted at different levels of accreditations.

AI can be used in many ways for teaching and learning. These include:

- Automated research summaries
- Citation and reference management
- Essay and report writing assistance
- Language and grammar enhancement
- Plagiarism detection
- Speech-to-text transcription
- Data analysis and visualisation
- Personalised learning material
- AI-generated quiz and exam questions
- Knowledge graphs and concept mapping
- Adaptive learning platforms
- AI-powered translation
- Automated peer review and feedback
- Content recommendation
- Code generation for research

AI can also help generate lecture notes, transcriptions of lectures, question banks with multiple types of questions, workshop manuals, and so on. AI based chatbots can add to classical teaching practices and can be used by research scholars for literature reviews, finding problem statements from research papers, extraction of future work, and so on.

Several LMS (learning management system) platforms and portals are available for online teaching and academic practices (Table 1). These come with a range of features including online attendance, student evaluation, question papers, data reports, video conferencing, and related activities.